

Project 1

Identifying and Tracking the Eight IK Branches of the PUMA 560

Format: Group of two students; simulation-based project.

Prerequisites: Rigid transforms, D–H modelling, analytical and numerical inverse kinematics, Jacobians, singularities, and path continuation.

Motivation and historical context

A Cartesian path specifies where a robot’s tool should move, but does not uniquely specify the robot’s posture. Although the PUMA 560 has six revolute joints and is nonredundant for a full-pose task, a regular target can admit up to eight isolated inverse-kinematic (IK) solutions. These postures reproduce the same tool pose but can have different joint-limit margins, singularity proximity, and joint travel along a subsequent path. Understanding their identity and continuation is therefore essential to choosing an initial posture that can execute the whole motion. Pointwise reachability alone does not establish path feasibility.

The PUMA family grew out of Victor Scheinman’s electrically driven robot-arm work and was developed at Unimation for industrial assembly in the late 1970s. The Smithsonian dates its Unimate PUMA 500 to 1978; the Computer History Museum describes the family’s use in automobile assembly and other industrial tasks.¹ The PUMA 560 became a widely studied example of a *wrist-partitioned 6R* manipulator: three arm joints position a wrist center, while three concurrent wrist axes determine orientation. This spherical-wrist geometry permits the full-pose IK problem to be separated into arm and wrist subproblems, yielding the familiar shoulder, elbow, and wrist choices.²

Six-axis arms remain relevant to industrial assembly and handling; current assembly applications include those documented by Stäubli.³ The planning questions studied here also arise when a tool must follow a process contour with a prescribed orientation. The PUMA provides a tractable model for studying these questions; its eight-class structure must not be assumed to apply to every industrial 6R geometry.

Task

Overall objectives. Build a consistent kinematic description of the PUMA 560, identify its eight regular IK classes, and evaluate whether this classification helps disambiguate numerical solutions near singularities. Then determine how the initial IK class affects the ability to follow a prescribed Cartesian pose path.

- **Understand and validate the model.** Inspect the supplied URDF: link/joint tree, joint origins and axes, limits, fixed transforms, and visual mesh origins and scales. Check link-name and mesh-path consistency when importing it. Establish a D–H table and document joint-angle offsets and base/tool conventions. Compare D–H forward kinematics with the URDF-based model at several configurations using position and orientation residuals. Explain why moving an STL visual origin does not change kinematics.

¹Smithsonian: Unimate PUMA 500; Computer History Museum: AI & Robotics timeline.

²Peter Corke, Robotics Toolbox: analytical spherical-wrist inverse kinematics.

³Stäubli: robotic assembly applications.

- **Recover and study eight IK solutions.** Use a regular full-pose target admitting eight mathematical solutions. Obtain a candidate target by applying forward kinematics to a regular configuration and checking its IK count. Reuse or adapt a validated analytical spherical-wrist IK implementation. Visualize every solution and verify pose closure. Distinguish mathematical solutions from those admitted by the chosen joint limits; record angle periodicity and remove equivalent duplicates.
- **Develop a geometric classifier.** Assign shoulder (left/right), elbow (up/down), and wrist (flip/non-flip) labels. Derive identification rules consistent with your D–H conventions and relate them to the factors of your model’s Jacobian determinant:

$$\det J(q) = K f_S(q) f_E(q) f_W(q), \quad (S, E, W) = (\operatorname{sgn} f_S, \operatorname{sgn} f_E, \operatorname{sgn} f_W).$$

Explain the mapping between signs and geometric names. Produce an illustrated table of the eight classes. Test identification after shuffling the solver’s output order, and explain why the sign of $\det J$ alone is insufficient.

- **Test classification near singularities.** Construct short approaches to shoulder, elbow, and wrist singularities. Monitor the corresponding factors and the smallest singular value of the Jacobian, documenting the scaling used to combine linear and angular quantities. Perturb the target poses slightly and compare geometric classification with nearest-configuration matching, with and without trajectory history. Examine which distinctions disappear or become numerically uncertain. Define tolerances for reporting an ambiguous label instead of forcing a sign. At a wrist singularity, explain the coupled freedom of the two outer wrist angles. Classification can preserve useful identity near a singularity; it cannot restore uniqueness when distinct regular branches cease to be distinguishable.
- **Continue each admissible initial class along a path.** Prepare three short paths: a regular case, a joint-limit case, and a near-singularity case. Specify both position and orientation. From each valid starting IK, continue its solution and retain its class history. Compare a nearest-solution baseline with selection informed by classification and continuation. Do not permit arbitrary jumps between classes: for this PUMA model, a continuous change of a separator sign requires a singular crossing.
- **Validate and interpret the trajectories.** Compare completed path fraction, joint travel, minimum joint-limit margin, singularity proximity, and label consistency. Resolve angle representatives consistently with the limits. Refine the path sampling near problematic transitions and re-solve the prescribed intermediate poses; feasible endpoints alone do not validate a connection. Explain failures and distinguish numerical ambiguity, joint-limit violations, and actual singularities. A sampled study is evidence, not a proof of continuous-path feasibility.

Provided materials and implementation

The PUMA 560 robot description (.urdf) and link meshes (.stl) will be provided to the students. Use a robotics library or your own validated tools to inspect frames and axes, verify D–H parameters, visualize IK postures, and animate paths. Confirm joint offsets and base/tool transforms when importing the supplied model.

Use full-pose IK to study all eight classes along a position-and-orientation path. A wrist-center-only solver with fixed wrist joints does not enforce the prescribed tool orientation; complement it with the spherical-wrist orientation solution if using that decomposition. Visual inspection should accompany quantitative checks.

Expected outcome

Students should be able to read and interpret the robot's URDF, establish and validate its D–H parameters, explain wrist-partitioned IK, and identify the eight regular shoulder–elbow–wrist classes. They should understand how singularities affect those classes, when identification becomes ambiguous, and why an initial IK choice matters for motion planning.

Submit reproducible code and test poses/paths, an illustrated classification table, comparison plots, a concise 5–6-page report, and a short simulation demonstration. Explain both successful continuations and failures. A positive result showing that classification always outperforms the baseline is not required.